HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

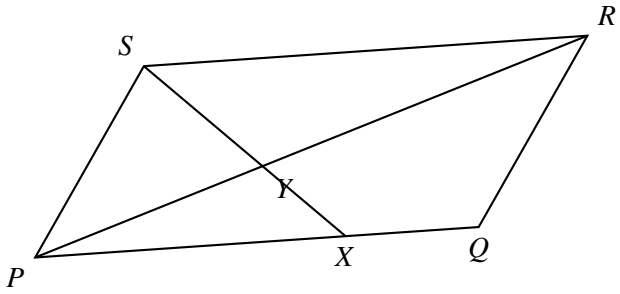
Student No.: 04 Name: _____ Marks: / 20

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. In the figure, $PQRS$ is a parallelogram. Let X be a point lying on PQ. Denote the point of intersection of PR and SX by Y. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are $32\,\text{cm}^2$ and $58\,\text{cm}^2$ respectively, then the area of $\triangle RSY$ is



A. $40\,\text{cm}^2$.

B. $50\,\text{cm}^2$.

C. $58\,\text{cm}^2$.

D. $72\,\text{cm}^2$.

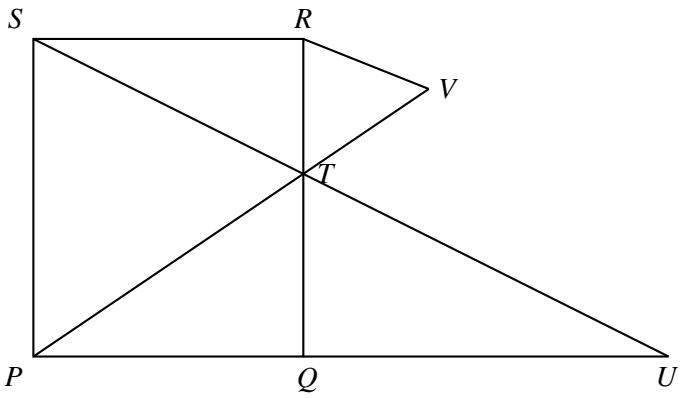
(DSE-2023 PII Q17)

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2. In the figure, $PQRS$ is a rectangle. Let T be a point lying on QR such that $\angle PTS = 90°$. PQ produced and ST produced meet at the point U. PT is produced to the point V such that $RT = RV$. Which of the following must be true?



A. $RV \parallel ST$

B. $\angle PTQ = \angle RTS$

C. $\triangle PST \sim \triangle UTQ$

D. $\triangle PQT \cong \triangle TRS$

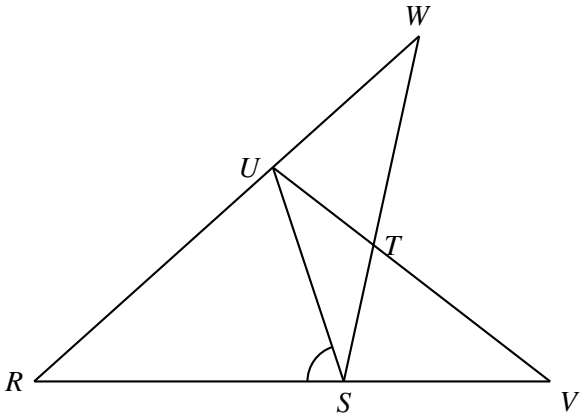
(DSE-2023 PII Q21)

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3. The figure shows the cyclic quadrilateral $RSTU$, where $ST = TU$. RS produced and UT produced meet at the point V while RU produced and ST produced meet at the point W. If $\angle RWS = 32^\circ$ and $\angle RVU = 48^\circ$, then $\angle RSU =$



A. 65°.

B. 73°.

C. 80°.

D. 82°.

(DSE-2023 PII Q22)

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4. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

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5. A box contains five cards numbered 1, 2, 3, 4 and 5 respectively while another box contains four cards numbered 6, 7, 8 and 9 respectively. If a number is randomly drawn from each box, find the probability that the product of the two numbers drawn is divisible by 4.

A. $\frac{1}{5}$

B. $\frac{3}{10}$

C. $\frac{7}{20}$

D. $\frac{9}{20}$

(DSE-2023 PII Q28)

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6. The L.C.M. of a^4b^2c, a^3b^4c and $a^2b^5c^2$ is

A. a^2b^2c.

B. $a^2b^2c^2$.

C. a^4b^5c.

D. $a^4b^5c^2$.

(DSE-2023 PII Q32)

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7. Consider the following system of inequalities:

$$\begin{cases} x+3 \geq 0 \\ 2x+3y-12 \leq 0 \\ 5x-3y+12 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the range of values of β such that the greatest value of $\beta x+6y$ is 24, where (x, y) is a point lying in D.

A. $\beta \leq -10$

B. $\beta \geq -10$

C. $\beta \leq 4$

D. $\beta \geq 4$

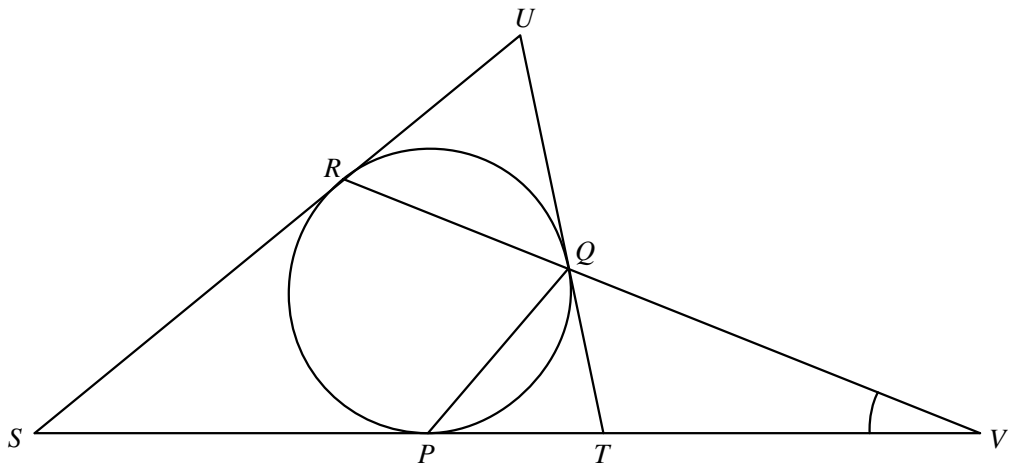
(DSE-2023 PII Q37)

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8. In the figure, P, Q and R are points lying on a circle. ST, TU and SU are the tangents to the circle at P, Q and R respectively. RQ produced and ST produced meet at the point V. If $\angle PSR = 34°$ and $\angle QPT = 46°$, then $\angle PVQ =$



A. $17°$.

B. $22°$.

C. $27°$.

D. $28°$.

(DSE-2023 PII Q38)

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9. When a boy throws a dart, the probability that he hits the target is 0.6 . If this boy throws the dart 4 times, find the probability that he hits the target at least 2 times.

A. 0.5248

B. 0.7056

C. 0.8208

D. 0.8464

(DSE-2023 PII Q43)

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10. It is given that n is an integer. Let u, v and w be the standard deviation, the median and the range of the group of numbers $\{1-9n, 3-9n, 4-9n, 5-9n, 7-9n\}$ respectively. Which of the following must be true?

I. $u = 2$

II. $v < 4$

III. $w > 6$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q45)

I can finish _____ questions

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